

Singularity-Free Low-Complexity Fault-Tolerant Prescribed Performance Control for Spacecraft Attitude Stabilization

Jin-Xi Zhang¹, *Member, IEEE*, Yun-Qi Liu, and Tianyou Chai², *Life Fellow, IEEE*

Abstract—This paper is concerned with the problem of fault-tolerant prescribed performance attitude stabilization for spacecraft under model uncertainties and actuator failures. In most of the existing control designs for spacecraft described by the unit quaternion, the possible singularity issue of the virtual control coefficient matrix is neglected such that the controllability of the attitude subsystem cannot be warranted throughout. On the other hand, the related works depend on complex algorithms of approximation, estimation or diagnosis to deal with unknown system dynamics. In this paper, a singularity-free low-complexity fault-tolerant prescribed performance control (PPC) strategy is put forward. To exclude the singularity issue, an initialization principle of the performance envelope is devised. On this basis, a static PPC law is developed, without parameter identification, function approximation, disturbance estimation, failure detection, failure isolation, failure estimation. In place of the classical Lyapunov stability theory, a unified performance analysis framework based on proof by contradiction and the barrier Lyapunov function is constructed. It not only turns out attitude stabilization with the preassigned settling time and accuracy whenever the actuator failures happen, but also discloses the intrinsic robustness of the control system versus actuator failures and model uncertainties. A comparative study on a rigid spacecraft is performed to demonstrate the validity and advantage of our approach.

Note to Practitioners—Attitude stabilization is a fundamental element for the spacecraft to accomplish various important missions. A mainstream method to model the spacecraft attitude is the unit quaternion. In this way, the control distribution matrix of the attitude subsystem becomes singular under certain attitude, which causes the loss of controllability. This problem is neglected by the majority of the existing attitude stabilization

methods. On the other side, the harsh space environment with high-intensity radiation, extremely low temperature and dust and fine particles gives rise to actuator failures in the spacecraft control system. Whilst the existing fault-tolerant control methods are capable of accommodating the actuator failures ultimately, the control performance during the post-failure transient stage is inevitably degraded. This problem is solved by the fault-tolerant prescribed performance control method, whereas the control algorithm shows a high complexity due to the dependence of function approximation or parameter identification. Motivated by the above observation, this paper presents a singularity-free low-complexity fault-tolerant prescribed performance control approach for the spacecraft with unknown dynamics and actuator failure. It achieves fast accurate attitude stabilization even in the faulty case.

Index Terms—Fault-tolerant control, low complexity, predefined performance, singularity-free control, spacecraft.

I. INTRODUCTION

SPACECRAFT is an important vehicle which travels in outer space to accomplish a variety of missions and purposes such as communications [1], earth observation [2], meteorology [3], navigation [4], space colonization [5], planetary exploration and transportation of humans and cargo. One of the criteria to make sure the spacecraft capable of completing its mission is the ability to control precisely the spacecraft orientation and attitude. Attitude control is the maintenance of prescribed desired attitude within a given tolerance [6], for example attitude stabilization as a fundamental control task. However, the differential equation governing the attitude motion is inherently nonlinear and highly coupled, and the attitude dynamics is inevitably affected by inertia uncertainties and external disturbances. Therefore, accurate attitude control of spacecraft is nontrivial. On the other hand, the harsh space environment with high-intensity radiation, extremely low temperature, dust and fine particles, etc, gives rise to faults of spacecraft control systems. Actuator failures are a fatal threat for the spacecraft, as its motion is changed straightforward by false control actions, which leads to performance degradation, instability or even a catastrophic accident. Fortunately, for the attitude control subsystem of the spacecraft, the redundant actuators offer possibility of accommodating actuator failures.

In the early results related on attitude stabilization of spacecraft [7], [8], [9], [10], [11], [12], it was only guaranteed that the stabilization errors exponentially converge to a neighborhood of zero whose size is unknown beforehand. To

Received 19 October 2024; revised 12 January 2025; accepted 5 May 2025. Date of publication 13 May 2025; date of current version 23 May 2025. This article was recommended for publication by Associate Editor C. Zhang and Editor C. Seatzu upon evaluation of the reviewers' comments. This work was supported in part by the National Natural Science Foundation of China under Grant 62473089, in part by the Research Program of Liaoning Liaohe Laboratory under Grant LLL23ZZ-05-01, in part by the 111 Project 2.0 of China under Grant B08015, in part by the National Key Research and Development Program of China under Grant 2022YFB3305905, in part by Xingliao Talent Program of Liaoning Province of China under Grant XLYC2203130, in part by the Natural Science Foundation of Liaoning Province of China under Grant 2023JH26/10200011 and Grant 2024JH3/10200012, in part by the Open Research Project of the State Key Laboratory of Industrial Control Technology of China under Grant ICT2024B12, and in part by the Fundamental Research Funds for the Central Universities of China under Grant N2424016 and Grant N25YJS002. (*Corresponding author: Tianyou Chai.*)

The authors are with the State Key Laboratory of Synthetical Automation for Process Industries, Northeastern University, Shenyang 110819, China (e-mail: zhangjx@mail.neu.edu.cn; liuyunqi@stumail.neu.edu.cn; tychai@mail.neu.edu.cn).

Digital Object Identifier 10.1109/TASE.2025.3569566

1558-3783 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on November 21, 2025 at 14:00:49 UTC from IEEE Xplore. Restrictions apply.

accelerate convergence, finite-time attitude stabilization was accomplished [13], [14]. Further, to relax the dependence of the settling time on the initial condition, fixed-time control was developed [15], [16], [17]. Nonetheless, the settling time and the accuracy cannot be quantitatively prespecified. Recently, the prescribed performance control (PPC) method was applied to spacecraft [18], [19], [20], [21], [22], [23], [24], [25], which ensures the evolutions of stabilization errors inside the preselected performance funnels such that the errors converge to a predefined accuracy bound with the rate no less than a preset value. However, it should be noted that gain adaptation [19], sliding mode [20], [21], [25], reinforcement learning [22] and disturbance estimation [22], [23] are adopted to deal with unknown nonlinearities as well as disturbances. As a cost, a large number of variables or parameters have to be updated online, which significantly increases the complexity and computational burden of the control algorithm, as pointed out by Reference [26].

On the other hand, the aforementioned solutions work for spacecraft in the fault-free case [7], [9], [10], [11], [12], [13], [14], [15], [16], [17], [19], [20], [23]. To accommodate actuator failures, an intelligent fault-tolerant control (FTC) law was designed [27], with the help of a trained neural network; a modified fault diagnosis unit was devised [28]; an extended observer was constructed to estimate and compensate for the synthetic uncertainties induced by actuator failures [29]; a fault-tolerant fuzzy switching controller was developed [30]; an adaptive mechanism was employed to estimate the bounds of actuator failures, disturbances and model uncertainties [31]. Nonetheless, only uniform ultimate boundedness of stabilization errors was guaranteed [27], [28], [29]. Although finite-time attitude stabilization was achieved [30], [31], the control accuracy still fails to be predefined. Moreover, post-failure performance is inevitably degraded [27], [28], [29], [30], [31], e.g., a large overshoot of the stabilization error in the post-fault phase which cannot be quantitatively regulated.

On the purpose of maintaining the expected performance of the faulty plant, a pair of fault-tolerant PPC strategies [32], [33] for spacecraft with actuator failures were developed in recent years. They both adopt the classical Lyapunov stability theory to analyze and design controllers. A Nussbaum gain-based fuzzy control scheme was developed [32], in which fuzzy logic systems delicate to accommodating model perturbations and disturbances, and Nussbaum-type functions devote to compensating for the loss of effectiveness of the actuators. A low-pass filter-based adaptive control strategy was put forward [33], in which the adaptive rule is utilized to estimate the bounds of the unknown nonlinearities and actuator failures, and the low-pass filter is deployed to avoid calculations of virtual control signals. As mentioned above, the resulting controller may be excessively complex as a result of approximation or estimation. Moreover, the approximation-based controller always ensures semi-global stability of the system.

Last but not least, the singularity issue of the attitude control system was not addressed in the aforesaid works [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [27], [28], [29], [30], [31],

[32]. In the kinematic equations of the spacecraft, the virtual control coefficient matrix depends on the unit quaternion of the spacecraft attitude. Under certain attitudes, the virtual control coefficient matrix becomes singular, such that the attitude subsystem is not controllable. This issue was neglected in most of the related studies; only an assumption on the singularity-free case was made [19], [31] with no verification of its rationality; an intuitive idea to avoid singularity by adapting the performance function was mentioned [33], however, no specific procedure was given. Until now, the problem of how to evade singularity to warrant controllability of the spacecraft with a rigorous theoretical insurance still remains open.

It is noteworthy that an approximation-free fault-tolerant PPC design for attitude tracking of the rigid spacecraft was reported recently [34]. However, the plant was assumed to be not only fully actuated, which circumvents the difficulty in dealing with the over-actuated problem in practice, but also well controllable without any singularity issue.

Motivated by the above observation, a low-complexity singularity-free fault-tolerant PPC strategy for the spacecraft with actuator failures, model uncertainties and external disturbances is proposed in this paper for the first time. It stabilizes the spacecraft attitude with the predefined time and accuracy. Its novelty and superiority with respect to the existing methods are summarized as follows.

- 1) An initialization principle of the performance envelopes is devised to exclude the possible singularity issue of the attitude control system [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [27], [28], [29], [30], [31], [32], [34].
- 2) Instead of compensation [23], [27], [28] or feedback linearization [32], [33], the unknown dynamics in the closed-loop system is addressed in a robust control manner. The inherent robustness of the controller is exploited by a unified performance analysis framework based on proof by contradiction and the barrier Lyapunov function (BLF).
- 3) The resulting controller is significantly simple, without sliding modes [20], [21], [31], [35], learning algorithms [22], neural networks [22], [27], [36], fuzzy logic systems [17], [30], [32], command filters [10], [33], adaptive observers [22], [23], [29], [37], adaptive mechanisms [10], [18], [27], [31], Nussbaum functions [18], [32], switching rules [30], gain adaption [34], [38], fault estimation [29], [31] or fault diagnosis [28].

The rest of the paper is organised as follows. Section II formulates the problem under consideration. Section III presents the controller development. Section IV gives the theoretical analysis to validate its effectiveness. In Section V, a simulation study is conducted to illustrate the theoretical result. In the end, Section VI concludes this paper.

II. PROBLEM STATEMENT

A. System Model

The kinematics and dynamics of the spacecraft under consideration are written by [5], [25], [27], [28], [29],

[31], and [32]

$$\begin{cases} \dot{q}_0 = -\frac{1}{2}q_v^T \omega, \\ \dot{q}_v = \frac{1}{2}(q_v^\times + q_0 I_3) \omega, \\ J\dot{\omega} = -\omega^\times J\omega + D\tau + d(t), \end{cases} \quad (1)$$

where $q_0 \in \mathbb{R}$ and $q_v = [q_1, q_2, q_3]^T \in \mathbb{R}^3$ constitute the unit quaternion which describes the spacecraft attitude; $\omega = [\omega_1, \omega_2, \omega_3]^T \in \mathbb{R}^3$ is a vector of the angular velocities of the spacecraft; $I_3 \in \mathbb{R}^{3 \times 3}$ is an identity matrix; $J \in \mathbb{R}^{3 \times 3}$ is the symmetric and positive inertia matrix; $\tau = [\tau_1, \dots, \tau_n]^T \in \mathbb{R}^n$ is the vector of the control torques produced by the reaction wheels [28] with $n > 3$; $D \in \mathbb{R}^{3 \times n}$ is the reaction wheel distribution matrix; $d(t) \in \mathbb{R}^3$ represents the external disturbances; $q_v^\times \in \mathbb{R}^{3 \times 3}$ and $\omega^\times \in \mathbb{R}^{3 \times 3}$ are the skew-symmetric matrices defined by

$$q_v^\times = \begin{bmatrix} 0 & -q_3 & q_2 \\ q_3 & 0 & -q_1 \\ -q_2 & q_1 & 0 \end{bmatrix},$$

$$\omega^\times = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}.$$

Property 1: The unit quaternion satisfies $q_0^2 + q_v^T q_v = 1$ [27], [30], [33], [39].

It is observed from (1) that τ regulates ω which regulates q_v which determines q_0 by Property 1. To ensure the controllability of the ω -subsystem, D should be with full row rank; to warrant the controllability of the q_v -subsystem, $(q_v^\times + q_0 I_3)$ ought to be nonsingular.

Assumption 1: The external disturbances are bounded but unknown [27], [28], [29], [31], [32].

Assumption 2: There holds $q_0(0) \neq 0$ [19], [31], [33].

B. Actuator Failures

The actuator failures of (1) are described by [27], [28], [31], and [33]

$$\tau_i(t) = \begin{cases} u_i(t), & \text{if } t < t_i, \\ \delta_i, & \text{otherwise,} \end{cases} \quad (2)$$

where $i \in \{1, \dots, n\}$; $u_i(t)$ denotes the i -th control signal to be designed below; $t_i \geq 0$ indicates the time instant at which the i -th actuator fails; δ_i is a constant which represents the stuck fault, e.g., $\delta_i = 0$ or $\delta_i = u_i(t_i)$.

Assumption 3: At least three reaction wheels keep healthy [25], [27], [28], [29], [31], [32], [33], [35].

Based on (2), the input vector of (1) is rewritten in the form of

$$\tau = A \cdot u + B \triangleq \begin{bmatrix} a_1 & & \\ & \ddots & \\ & & a_n \end{bmatrix} \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}, \quad (3)$$

where $a_i = 1$ and $b_i = 0$ indicate that the i -th actuator is failure-free; $a_i = 0$ and $b_i = \delta_i$ indicate that the i -th actuator undergoes the failure. Substituting (3) into (1) leads to

$$\begin{cases} \dot{q}_v = G_1(q)\omega, \\ \dot{\omega} = F(\omega) + G_2 u + E(t), \end{cases} \quad (4)$$

where $F(\omega) = -J^{-1}\omega^\times J\omega$, $G_2 = J^{-1}DA$, $E(t) = J^{-1}DB + J^{-1}d(t)$ and

$$G_1(q) = \frac{1}{2}(q_v^\times + q_0 I_3) = \frac{1}{2} \begin{bmatrix} q_0 & -q_3 & q_2 \\ q_3 & q_0 & -q_1 \\ -q_2 & q_1 & q_0 \end{bmatrix}. \quad (5)$$

It is seen from (4) that for the faulty attitude system, u regulates ω which regulates q_v which determines q_0 . The main difference between (1) and (4) lies in G_2 . To ensure the controllability of the ω -subsystem, G_2 should be with full row rank.

Assumption 4 ([40], [41]): The matrix G_2 can be decomposed into

$$G_2 = G_0 \cdot M, \quad (6)$$

where $G_0 \in \mathbb{R}^{3 \times n}$ is known with full row rank; $M \in \mathbb{R}^{n \times n}$ is unknown and unnecessarily symmetric; $G_0(M^T + M)G_0^T$ is positive definite.

Substituting (6) into (4) yields

$$\begin{cases} \dot{q}_v = G_1(q)\omega, \\ \dot{\omega} = F(\omega) + G_0 M u + E(t). \end{cases} \quad (7)$$

Assumption 4 means that $G_0 M$ is with full row rank. Therefore, the controllability of (7) (i.e., (1) or (4)) is determined by $G_1(q)$ which should be always nonsingular.

Remark 1: In some related works, the uncertainties in the system matrices [28], [31] and the disturbance bounds [8], [9], [31] are assumed to be known. In this study, the system matrices $F(\omega)$ and M and the disturbance vector $E(t)$ are allowed to be totally unknown; only G_0 and $G_1(q)$ need to be known for the control design. The partially available information of the system matrices is necessary for control of over-actuated systems [32], [33], [40].

C. Control Task

The control objective for (1) is to stabilize q_1, q_2 and q_3 to a small neighborhood around zero with the preassigned transient and steady-state performance:

$$|q_i(t)| < \underline{k}_{1i}, \quad t \geq T_i, \quad i = 1, 2, 3, \quad (8)$$

where $\underline{k}_{1i} > 0$ and $T_i > 0$ quantify the stabilization accuracy and the settling time, respectively. Theoretically, the performance indices are allowed to be arbitrarily set. In practical applications, however, they have to be dependent on the input constraint of the control system and the measurement accuracy of the sensors, respectively.

To sum up, the problem investigated in the present paper is stated as follows.

Problem 1: For the spacecraft in (1) with Property 1 subject to the actuator failures in (2), devise an FTC scheme to guarantee the performance specifications in (8) and the boundedness of the signals involved in the control system.

III. FAULT-TOLERANT PPC DESIGN

To cope with Problem 1, a fault-tolerant PPC approach together with an initialization principle of performance envelopes is developed below.

A. Performance Functions With an Initialization Principle

The control design begins with the performance design which should not only fulfill the performance requirement but also guarantee the control feasibility. To this end, we first construct a tuning function:

$$\phi_i(t) = \begin{cases} 1 - \sin\left(\frac{\pi}{2} \frac{t}{T_i}\right), & \text{if } t < T_i, \\ 0, & \text{otherwise,} \end{cases} \quad (9)$$

for $i = 1, 2, 3$. Then, based on (9), the control objective in (8) is formulated by

$$|q_i(t)| < k_{1i}(t) = (\bar{k}_{1i} - \underline{k}_{1i}) \phi_i(t) + \underline{k}_{1i}, \quad t \geq 0, \quad (10)$$

for $i = 1, 2, 3$, with

$$\bar{k}_{1i} = p_i + |q_i(0)|, \quad i = 1, 2, 3, \quad (11)$$

where p_i is a constant and should meet

$$0 < p_i < \frac{-m + \sqrt{m^2 + 3q_0^2(0)}}{3}, \quad (12)$$

with $m = |q_1(0)| + |q_2(0)| + |q_3(0)|$. One sees from the above design that three predefined-time and smoothly decaying function in (10) with (9) are generated, instead of the exponential performance functions [42], [43], [44], [45], [46]. After $t = T_i$, the attitude constraints in (10) become the performance requirements in (8). Moreover, (12) is feasible under Assumption 2. Based on the parameter selections in (11) with (12), the following initial condition holds:

$$|q_i(0)| < k_{1i}(0), \quad i = 1, 2, 3. \quad (13)$$

A key technical lemma is proposed as follows, which shows that the singularity of the kinematic matrix can be evaded by imposing the performance boundary on the spacecraft attitude.

Lemma 1: For any $t_f > 0$, if

$$|q_i(t)| < k_{1i}(t), \quad t < t_f, \quad i = 1, 2, 3, \quad (14)$$

then $G_1(q)$ is nonsingular on $[0, t_f]$.

Proof: It follows from (9)–(11) that

$$k_{1i}(t) < k_{1i}(0) = \bar{k}_{1i}, \quad i = 1, 2, 3, \quad t > 0. \quad (15)$$

Eq. (5) yields

$$\det(G_1(q)) = \frac{1}{2} q_0. \quad (16)$$

Property 1 shows

$$q_0^2 = 1 - (q_1^2 + q_2^2 + q_3^2). \quad (17)$$

In the case of (14), substituting (15) into (17) yields

$$q_0^2(t) > 1 - (\bar{k}_{11}^2 + \bar{k}_{12}^2 + \bar{k}_{13}^2), \quad t < t_f. \quad (18)$$

It follows from (11) that

$$\bar{k}_{11}^2 + \bar{k}_{12}^2 + \bar{k}_{13}^2 = \sum_{i=1}^3 p_i^2 + \sum_{i=1}^3 2p_i |q_i(0)| + 1 - q_0^2(0). \quad (19)$$

According to (12), there hold

$$\begin{aligned} \sum_{i=1}^3 p_i^2 &< \frac{(m - \sqrt{m^2 + 3q_0^2(0)})^2}{3} \\ &= \frac{1}{3} \left(2m^2 + 3q_0^2(0) - 2m \sqrt{m^2 + 3q_0^2(0)} \right), \end{aligned} \quad (20)$$

and

$$\begin{aligned} \sum_{i=1}^3 2p_i |q_i(0)| &< \frac{2m (\sqrt{m^2 + 3q_0^2(0)} - m)}{3} \\ &= \frac{1}{3} \left(-2m^2 + 2m \sqrt{m^2 + 3q_0^2(0)} \right). \end{aligned} \quad (21)$$

From (20) and (21), we have

$$\sum_{i=1}^3 p_i^2 + \sum_{i=1}^3 2p_i |q_i(0)| < q_0^2(0). \quad (22)$$

Substituting (22) into (19) yields

$$\bar{k}_{11}^2 + \bar{k}_{12}^2 + \bar{k}_{13}^2 < 1. \quad (23)$$

Note that $\bar{k}_{1i}$, $i = 1, 2, 3$, are constants. Then, let

$$\underline{k} = 1 - \bar{k}_{11}^2 - \bar{k}_{12}^2 - \bar{k}_{13}^2 > 0. \quad (24)$$

Substituting (23) with (24) into (18), we have

$$q_0^2(t) \geq \underline{k} > 0, \quad t < t_f. \quad (25)$$

Hence, $G_1(q)$ with $\det(G_1(q)) = \frac{1}{2} q_0(t)$ is nonsingular on $[0, t_f]$. ■

Remark 2: Lemma 1 indicates that under (14), not only the prescribed stabilization performance in (8) is met, but also the controllability of the q_v -subsystem in (7) is guaranteed. This is attributed to the skillful construction of the performance functions in (10) based on the initialization principle in (11) and (12). In accordance with this guidance, we next develop the controller to ensure that (14) holds for all $t \geq 0$.

B. Static PPC Law

The controller design follows the recursive procedure. On the purpose of (10), employ the following barrier functions:

$$\eta_{1i}(t) = \tan\left(\frac{\pi}{2} \cdot \frac{q_i(t)}{k_{1i}(t)}\right), \quad i = 1, 2, 3. \quad (26)$$

Such functions are commonly used to handle constraints. Let

$$\eta_1(t) = [\eta_{11}(t), \eta_{12}(t), \eta_{13}(t)]^T. \quad (27)$$

Regard the angular velocities, $\omega(t)$, as the intermediate control inputs of (1) to regulate the q_v -subsystem. The corresponding virtual control law is formed by

$$\alpha(t) = -c_1 G_1^{-1}(q) \eta_1(t) := [\alpha_1(t), \alpha_2(t), \alpha_3(t)]^T, \quad (28)$$

where $c_1 > 0$ is a self-designed parameter. The rest of the task is to steer $\omega(t)$ to track $\alpha(t)$. The resulting intermediate error vector is

$$e(t) = w(t) - \alpha(t) := [e_1(t), e_2(t), e_3(t)]^T. \quad (29)$$

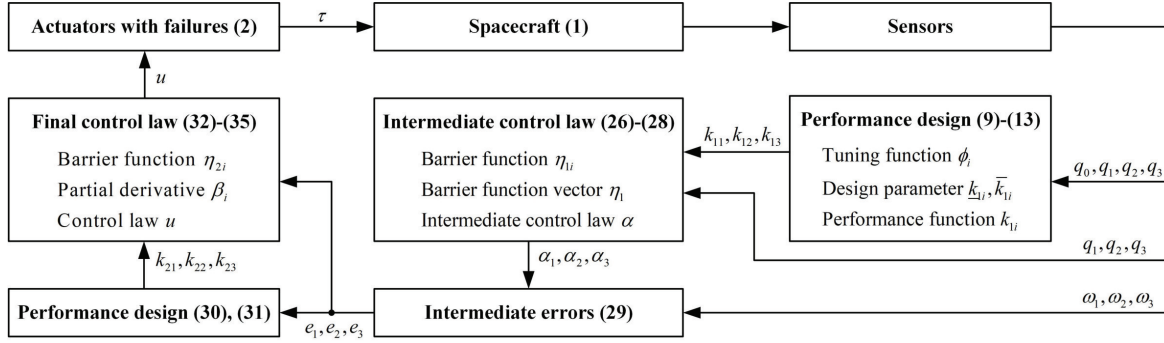


Fig. 1. A schematic of the proposed control approach.

To prescribe the stabilization performance of the intermediate errors, select the following performance functions:

$$k_{2i}(t) = (\bar{k}_{2i} - \underline{k}_{2i}) e^{-\mu_{2i}t} + \underline{k}_{2i}, \quad i = 1, 2, 3, \quad (30)$$

where $\bar{k}_{2i} > \underline{k}_{2i} > 0$ and $\mu_{2i} > 0$ are constants; $\underline{k}_{2i}$ and μ_{2i} are freely determined by the designer; $\bar{k}_{2i}$ should meet

$$|e_i(0)| < k_{2i}(0). \quad (31)$$

Herein, μ_{2i} and $\underline{k}_{2i}$ quantify the convergence speed and the stabilization accuracy of $e_i(t)$, respectively, $i = 1, 2, 3$. To this end, three barrier functions are adopted:

$$\eta_{2i}(t) = \tan\left(\frac{\pi}{2} \cdot \frac{e_i(t)}{k_{2i}(t)}\right), \quad i = 1, 2, 3. \quad (32)$$

Further, in order to deal with the inputs coupling of (1), the partial derivatives of (32) are involved:

$$\beta_i(t) = \frac{1}{k_{2i}(t)} \cdot \frac{1}{\cos^2\left(\frac{\pi}{2} \cdot \frac{e_i(t)}{k_{2i}(t)}\right)}, \quad i = 1, 2, 3. \quad (33)$$

For ease of notation, let

$$\eta_2(t) = \begin{bmatrix} \eta_{21}(t) \\ \eta_{22}(t) \\ \eta_{23}(t) \end{bmatrix}, \quad \beta(t) = \begin{bmatrix} \beta_1(t) \\ \beta_2(t) \\ \beta_3(t) \end{bmatrix}. \quad (34)$$

Based on Assumption 4 and (34), the final control law is obtained as follows

$$u(t) = -c_2 G_0^T \beta(t) \eta_2(t), \quad (35)$$

where $c_2 > 0$ is a free-design parameter. A schematic of the resulting control system is exhibited in Fig. 1.

Remark 3: Careful inspection of (9)–(12) and (26)–(35) reveals that the control law is independent of information about the system matrices $F(\omega)$ and M and the disturbance vector $E(t)$ in (7). Besides, it involves no derivatives of the intermediate control signals in (28). Nonetheless, no sliding modes [20], [21], [31], adaptive observers [22], [23], [29], command filters [10], [33], neural networks [22], [27], fuzzy logic systems [17], [30], [32], fault estimation [29], [31] or fault diagnosis [28] are employed. Therefore, a static controller is obtained, which is both less demanding and structurally simple.

IV. PERFORMANCE ANALYSIS

The theoretical result of the present paper is summarized below. Before that, a lemma is provided as follows, which manifests that $G_0 M G_0^T$ is a positive definite matrix.

Lemma 2: Let $g > 0$ be the minimum eigenvalue of $G_0(M + M^T)G_0^T$ in Assumption 4. For any $X \in \mathbb{R}^3$, there holds

$$X^T G_0 M G_0^T X \geq \frac{g}{2} \|X\|^2.$$

Proof: There is

$$G_0 M G_0^T = \frac{1}{2} G_0 (M + M^T) G_0^T + \frac{1}{2} G_0 (M - M^T) G_0^T.$$

We thus have

$$\begin{aligned} X^T G_0 M G_0^T X &= \frac{1}{2} X^T G_0 (M + M^T) G_0^T X \\ &\quad + \frac{1}{2} X^T G_0 (M - M^T) G_0^T X \\ &= \frac{1}{2} X^T G_0 (M + M^T) G_0^T X \\ &\quad + \frac{1}{2} X^T G_0 M G_0^T X - \frac{1}{2} X^T G_0 M^T G_0^T X. \end{aligned} \quad (36)$$

Since $X^T G_0 M^T G_0^T X$ is a scalar, there is

$$X^T G_0 M^T G_0^T X = (X^T G_0 M^T G_0^T X)^T = X^T G_0 M G_0^T X.$$

Then, (36) becomes

$$X G_0 M G_0^T X^T = \frac{1}{2} X G_0 (M + M^T) G_0^T X^T.$$

It follows from Assumption 4 that

$$X^T G_0 M G_0^T X \geq \frac{g}{2} \|X\|^2.$$

Another lemma is given as follows, which shows the quantitative relationship between $\|\beta(t)\eta_2(t)\|$ and $\|\eta_2(t)\|$.

Lemma 3: Let $p = \max\{\bar{k}_{21}, \bar{k}_{22}, \bar{k}_{23}\}$. Then, there holds

$$\|\beta(t)\eta_2(t)\| \geq \frac{1}{p} \|\eta_2(t)\|. \quad (37)$$

Proof: From (32)–(34), we have

$$\|\beta(t)\eta_2(t)\| = \sqrt{\sum_{i=1}^3 \frac{\eta_{2i}^2(t)}{k_{2i}^2(t) \cos^4\left(\frac{\pi}{2} \cdot \frac{e_i(t)}{k_{2i}(t)}\right)}}$$

$$\geq \sqrt{\sum_{i=1}^3 \frac{\eta_{2i}^2(t)}{k_i^2(t)}}. \quad (38)$$

It follows from (30) that

$$\frac{1}{k_{2i}(t)} \geq \frac{1}{k_{2i}(0)} \geq \frac{1}{p}, \quad i = 1, 2, 3. \quad (39)$$

Substituting (39) into (38) yields (37).■

Theorem 1: Problem 1 is addressed by the controller in (9)–(12) and (26)–(35), under Assumptions 1–4.

Proof: See Appendix.■

Remark 4: Based upon the classical Lyapunov stability theory for the analysis and design of the controller, the unknown dynamics of the closed-loop system need to be compensated for feedback linearization. For example, employ neural networks and adaptive estimators to counteract the effects of model uncertainties and actuator failures, respectively. In place of this traditional way, a unified performance analysis based on proof by contradiction and a BLF is carried out above. It reveals that under the variable constraints in (52), the unknown closed-loop dynamics in (58) with (56) and (75) with (72) is bounded, but the barrier function-based control signal is enlarged to infinity once the constrained variable tends to the performance boundary (see, e.g., (59)). This means that the control signals suffice to cancel out the unknown closed-loop dynamics, without the aid of auxiliary tools mentioned in Remark 3. Thus, our performance analysis discloses that the designed controller is inherently robust to the external disturbances, model uncertainties and actuator failures.

Remark 5: On the purpose of overshoot regulation of the spacecraft attitude, i.e.,

$$\begin{cases} -\underline{k}_{1i} < q_i(t) < k_{1i}(t), & t \geq 0, \text{ if } q_i(0) \geq 0, \\ -k_{1i}(t) < q_i(t) < \underline{k}_{1i}, & t \geq 0, \text{ if } q_i(0) < 0, \end{cases} \quad (40)$$

For $i = 1, 2, 3$, we first make the following transformation:

$$\begin{aligned} \hat{k}_{1i}(t) &= \frac{1}{2}(k_{1i}(t) + \underline{k}_{1i}), \\ \hat{q}_i(t) &= q_i(t) - \frac{1}{2}(k_{1i}(t) - \underline{k}_{1i}), \text{ if } q_i(0) \geq 0, \\ \hat{q}_i(t) &= q_i(t) - \frac{1}{2}(\underline{k}_{1i} - k_{1i}(t)), \text{ if } q_i(0) < 0. \end{aligned}$$

Then, the asymmetric constraint in (40) is transformed to

$$|\hat{q}_i(t)| < \hat{k}_{1i}(t), \quad t \geq 0, \quad i = 1, 2, 3.$$

Accordingly, the barrier functions in (26) are rewritten by

$$\eta_{1i}(t) = \tan\left(\frac{\pi}{2} \cdot \frac{\hat{q}_i(t)}{\hat{k}_{1i}(t)}\right), \quad i = 1, 2, 3.$$

This together with (27)–(35) in the control algorithm guarantees (40).

V. SIMULATION STUDY

Now, we conduct a comparative simulation on a rigid spacecraft [27] to illustrate the above theoretical result. The plant is described by (1) with $n = 6$, and the inertia matrix

J and the reaction wheel distribution matrix D are given by [27], [37]

$$J = \begin{bmatrix} 2.5 & 0.002 & 0.0076 \\ 0.002 & 2.6 & 0.01 \\ 0.0076 & 0.01 & 1.9 \end{bmatrix} \text{ kg} \cdot \text{m}^2, \quad (41)$$

$$D = \begin{bmatrix} 0.8 & -0.8 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.7 & -0.7 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.7 & -0.7 \end{bmatrix}. \quad (42)$$

Let the initial condition be

$$q_v(0) = [-0.1, 0.12, -0.13]^T,$$

$$q_0(0) = 0.9791,$$

$$\omega(0) = [0, 0, 0]^T \text{ rad/s}.$$

To evaluate the fault tolerance of the control system, the actuator failures are simulated by

$$\tau_1(t) = \begin{cases} u_1(t), & \text{if } t < 100s, \\ 0.03, & \text{otherwise,} \end{cases} \quad (43)$$

$$\tau_3(t) = \begin{cases} u_3(t), & \text{if } t < 150s, \\ 0, & \text{otherwise,} \end{cases} \quad (44)$$

$$\tau_5(t) = \begin{cases} u_5(t), & \text{if } t < 200s, \\ 0, & \text{otherwise.} \end{cases} \quad (45)$$

The rest of the actuators are always keeping healthy, i.e., $\tau_j(t) = u_j(t)$, $j = 2, 4, 6$. To test the robustness of the control system, the external disturbances are simulated by [37]

$$d(t) = \begin{bmatrix} 4 \sin(0.03t) + 2 \cos(0.1t) - 2 \\ -1.5 \sin(0.02t) + 3 \cos(0.05t) + 2 \\ 3 \sin(0.1t) - 8 \cos(0.04t) + 2 \end{bmatrix} \times 10^{-4} \text{ N} \cdot \text{m}. \quad (46)$$

As pointed out by Reference [37], $d(t)$ is sufficient to cover the actual disturbances in the space environment. One sees from (1) that the total input of the spacecraft is $\Gamma = D\tau$. From a practical point of view, Γ is constrained by [37]

$$|\Gamma_i(t)| \leq 0.05 \text{ N} \cdot \text{m}, \quad i = 1, 2, 3, \quad (47)$$

where Γ_i denotes the i -th element of Γ .

The control goal is attitude stabilization with the prescribed performance specifications:

$$|q_i(t)| < 0.025, \quad t \geq 155s, \quad i = 1, 2, 3. \quad (48)$$

According to Theorem 1, a static controller is obtained with $c_1 = 5$, $c_2 = 8$, $\bar{k}_{21} = 1.4$, $\bar{k}_{22} = \bar{k}_{23} = 2.5$, $\underline{k}_{21} = 1$, $\underline{k}_{22} = \underline{k}_{23} = 2$, $\mu_{2i} = 0.08$, $i = 1, 2, 3$, and

$$G_0 = \begin{bmatrix} 0.0012 & -0.0012 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.001 & -0.001 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.001 & -0.001 \end{bmatrix}.$$

The simulation is conducted under MATLAB Simulink with the 0.0 start time, the 300 stop time, the fixed-step type, the Bogacki-Shampine solver, the 0.001 fundamental sample time, the unconstrained periodic sample time, and the tasking mode that treats each discrete rate as a separate task.

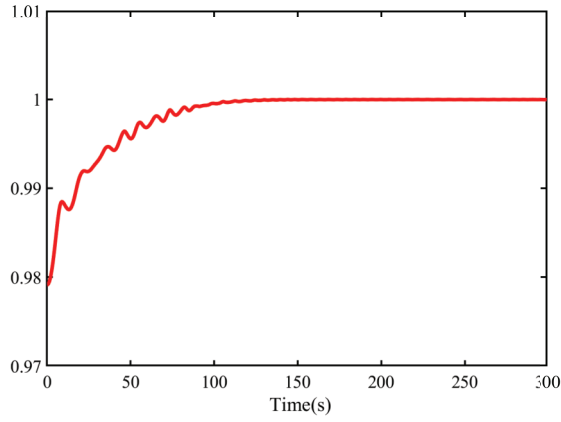
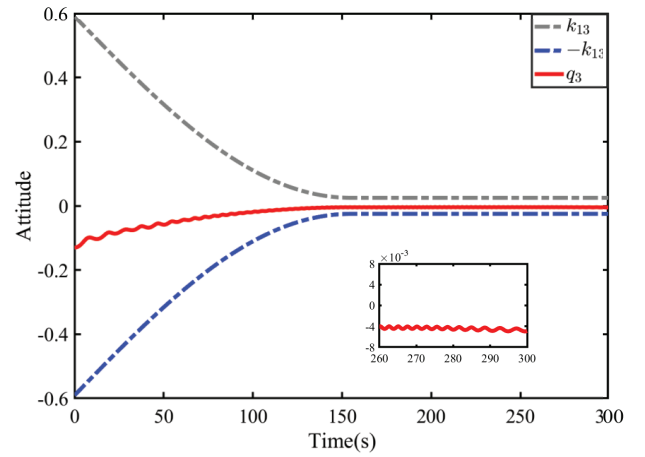
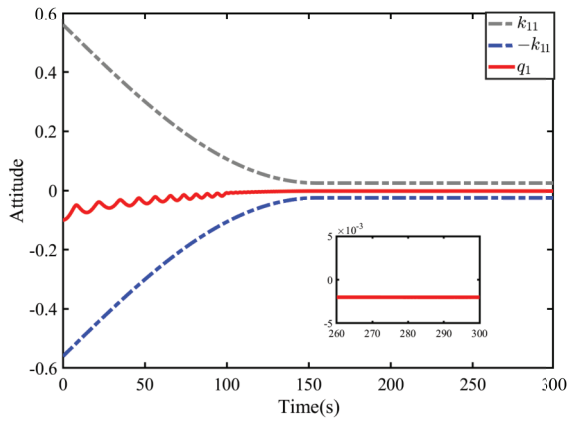
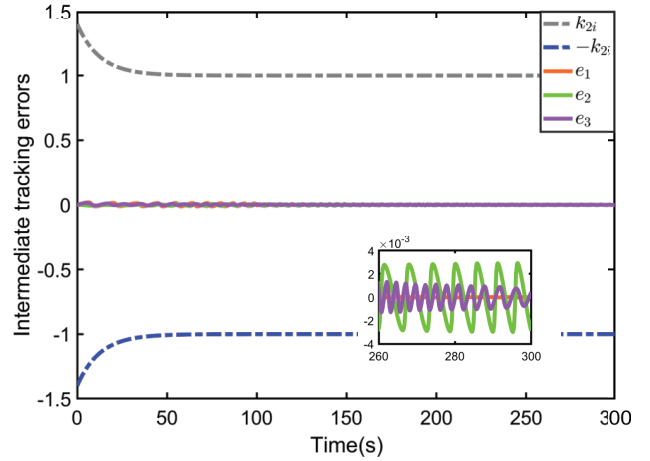
Fig. 2. The attitude q_0 .Fig. 5. The attitude q_3 .Fig. 3. The attitude q_1 .

Fig. 6. The intermediate tracking errors.

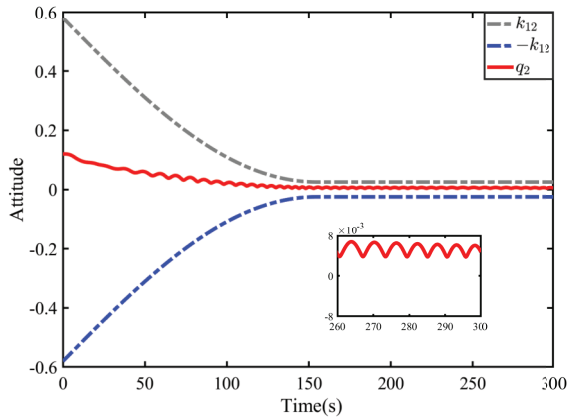
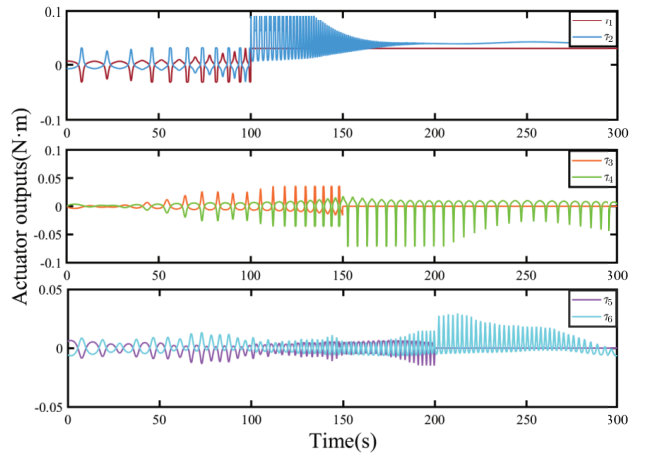
Fig. 4. The attitude q_2 .

Fig. 7. The actuator outputs.

Carry out simulation on the spacecraft with the above controller. The results are exhibited in Figures 2–8 and analyzed as follows. First, Fig. 2 shows that q_0 keeps away from zero, which ensures the controllability of the spacecraft. Figures 3–5 display that each attitude of the spacecraft evolves within the predefined performance funnel and ultimately converges to a small neighborhood of zero. Thus, the performance requirement in (48) is satisfied, despite the unknown model parameters in (41) and (42), the time-varying disturbances in

(46), and the multiple actuator failures in (43)–(45). Similarly, the intermediate errors plotted in Fig. 6 are confined inside the prescribed performance envelopes. Finally, it is observed from Figures 7 and 8 that the actuator outputs are bounded and the total inputs obey the prescribed constraints in (47),

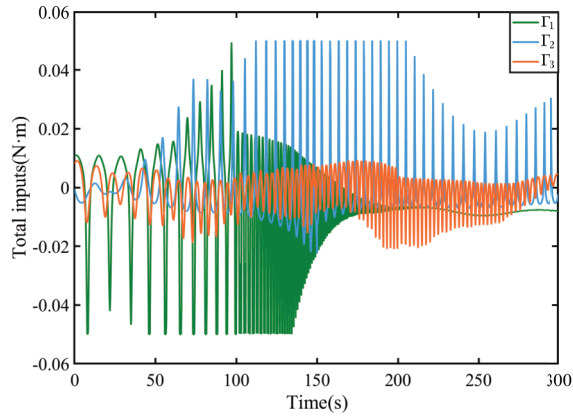


Fig. 8. The total inputs.

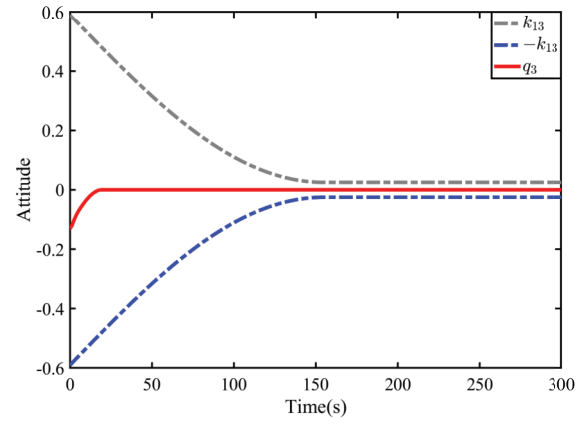
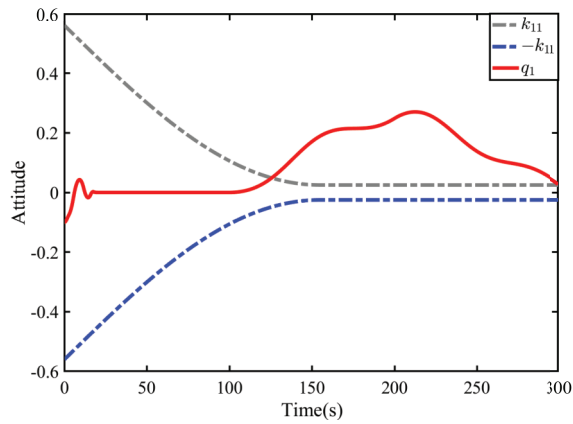
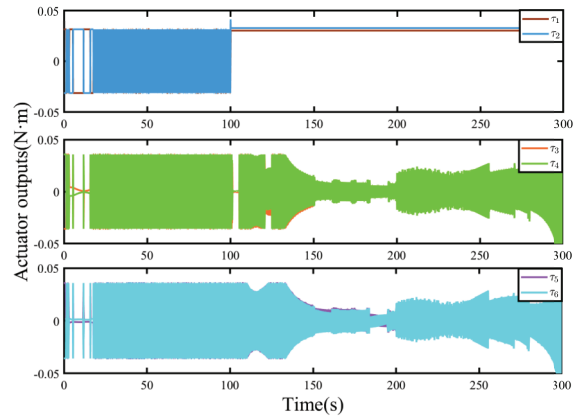
Fig. 11. The attitude q_3 by the comparative controller.Fig. 9. The attitude q_1 by the comparative controller.

Fig. 12. The actuator outputs by the comparative controller.

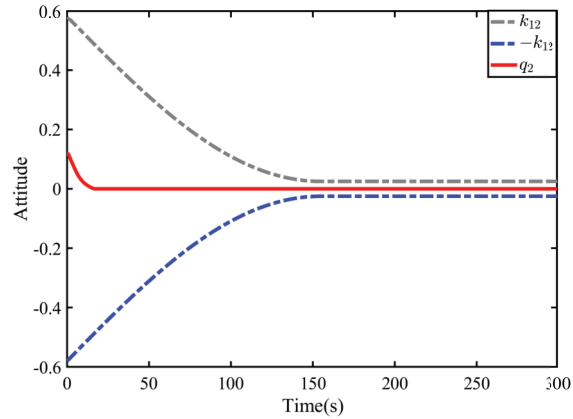
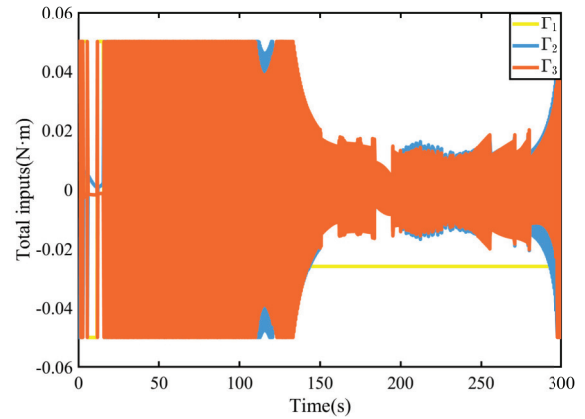
Fig. 10. The attitude q_2 by the comparative controller.

Fig. 13. The total inputs by the comparative controller.

respectively. Accordingly, the simulation results verify the feasibility of the controller.

For comparison, a fully adaptive-gain-based intelligent FTC scheme [27] is applied to the rigid spacecraft with the model parameters in (41) and (42), the time-varying disturbances in (46), the actuator failures in (43)–(45), and the input constraints in (47). The simulation results are depicted in Figures 9–13. It is shown in Figures 10 and 11 that the stabilization of q_2 and q_3 with the prescribed

performance specifications is achieved, despite the actuator failures. However, q_1 plotted in Fig. 9 evolves outside the performance funnel after $t = 120$ s caused by the actuator failures. It is displayed in Figures 12 and 13 that the actuator outputs are bounded and the total inputs fulfill the constraints. Nevertheless, both τ_i and Γ_j exhibit significant high-frequency oscillations between positive and negative, $i = 1, \dots, 6$, $j = 1, 2, 3$, which is not expected in practice and even prohibitive in certain applications.

Therefore, the comparative results illustrate the superiority of our approach in the fault tolerance and the control performance.

VI. CONCLUSION

This paper presents a singularity-free low-complexity fault-tolerant prescribed performance control approach for spacecraft with model uncertainties, external disturbances and actuator failures. It achieves attitude stabilization with the prespecified settling time and accuracy, whenever the actuator failures take place. Compared with the existing methods, our approach excludes the possible singularity issue of the virtual control coefficient matrix and hence rigorously guarantees the controllability of the attitude system. Moreover, the controller requires only partially available control distribution matrix and shows an extraordinary simplicity without adaption, identification, approximation, estimation, or diagnosis. The comparative simulation results of a rigid spacecraft demonstrate the theoretical findings. As future work, it is of interest to extend the proposed approach to more complex flight control systems, e.g., trajectory tracking of autonomous aerial vehicles.

APPENDIX

In Section III, a recursive constrained controller is devised. Now, we substantiate its feasibility in what follows, i.e., all the variables involved in the barrier functions are confined within the given envelopes:

$$\begin{cases} |q_i(t)| < k_{1i}(t), & i = 1, 2, 3, \quad t \geq 0, \\ |e_i(t)| < k_{2i}(t), & i = 1, 2, 3, \quad t \geq 0. \end{cases} \quad (49)$$

In order to establish this claim, the proof by contradiction inspired by [38], [47], and [48] is adopted, which is more intuitive compared with the classical analysis based on barrier Lyapunov functions. Recalling (13) and (31), we know that (49) holds at $t = 0$. According to (7) and (9)–(12), $q_i(t)$, $\omega_i(t)$ and $k_{1i}(t)$ are continuous, $i = 1, 2, 3$. Hence, $\eta_{1i}(t)$ in (34) and $\alpha_i(t)$ in (28) are continuous, if $|q_i(t)| < k_i(t)$, $i = 1, 2, 3$. Further, $e_i(t)$, $i = 1, 2, 3$, in (29) are continuous under the same condition. These findings indicate a necessary condition for the violation of (49): at least one constrained variable tends to the prescribed boundary. That is to say, if (49) is violated, then there should exist $\hat{t} > 0$ so that

$$\lim_{t \rightarrow \hat{t}} |q_i(t)| = \lim_{t \rightarrow \hat{t}} k_{1i}(t), \quad i \in \{1, 2, 3\}, \quad (50)$$

or

$$\lim_{t \rightarrow \hat{t}} |e_i(t)| = \lim_{t \rightarrow \hat{t}} k_{2i}(t), \quad i \in \{1, 2, 3\}, \quad (51)$$

and

$$|q_i(t)| < k_{1i}(t), \quad |e_i(t)| < k_{2i}(t), \quad i = 1, 2, 3, \quad t < \hat{t}. \quad (52)$$

Note that since (49) is defined on $[0, +\infty)$, $\hat{t}$ may either be a finite time instant or tend to infinity. Next, we suppose (50) or (51) with (52) and enumerate each case in (50) or (51) for verification.

Case 1: At the outset, consider (50), for which a necessary condition is

$$\lim_{t \rightarrow \hat{t}} \frac{d|q_i(t)|}{dt} \geq \lim_{t \rightarrow \hat{t}} \dot{k}_{1i}(t), \quad i \in \{1, 2, 3\}. \quad (53)$$

As a matter of fact, $\dot{k}_{1i}(t)$, $i = 1, 2, 3$, have lower bounds. From (9) and (10), we have

$$\dot{k}_{1i}(t) = (\underline{k}_{1i} - \bar{k}_{1i}) \left(\cos \left(\frac{\pi}{2} \cdot \frac{t}{T_i} \right) \right) \frac{\pi}{2T_i}, \quad i = 1, 2, 3,$$

for $i = 1, 2, 3$. There thus holds

$$\frac{\pi (\underline{k}_{1i} - \bar{k}_{1i})}{2T_i} \leq \dot{k}_{1i}(t) \leq 0, \quad i = 1, 2, 3.$$

Substituting it into (53) shows

$$\lim_{t \rightarrow \hat{t}} \frac{d|q_i(t)|}{dt} > -\infty, \quad i \in \{1, 2, 3\}. \quad (54)$$

Next, we analyze the dynamics of $q_i(t)$, $i = 1, 2, 3$. Rewrite (29) as follows

$$w(t) = e(t) + \alpha(t). \quad (55)$$

Substituting (55) with (28) into the second equation in (7) yields

$$\begin{aligned} \dot{q}_v(t) &= G_1(q)(e(t) - c_1 G_1^{-1}(q) \eta_1(t)) \\ &= G_1(q) \cdot e(t) - c_1 \eta_1(t). \end{aligned}$$

Based on (27), split $\dot{q}_v(t)$ as follows

$$\dot{q}_i(t) = \hat{G}_i(t) - c_1 \eta_{1i}(t), \quad i = 1, 2, 3, \quad (56)$$

where $\hat{G}_i(t)$ denotes the i th element of $(G_1(q) \cdot e(t))$. Then, we obtain

$$\lim_{t \rightarrow \hat{t}} \frac{d|q_i(t)|}{dt} = \lim_{t \rightarrow \hat{t}} \text{sign}(q_i) (\hat{G}_i(t) - c_1 \eta_{1i}(t)), \quad (57)$$

for $i = 1, 2, 3$. Proceed with discussing the values of the two parts in (57), respectively. On the one hand, we know from (52) that

$$|\hat{G}_i(t)| < \infty, \quad t < \hat{t}, \quad i = 1, 2, 3. \quad (58)$$

On the other hand, according to (26) and (50), we have

$$\lim_{t \rightarrow \hat{t}} \text{sign}(q_i) \eta_{1i}(t) = +\infty, \quad i \in \{1, 2, 3\}. \quad (59)$$

Substituting (59) and (58) into (57) yields

$$\lim_{t \rightarrow \hat{t}} \frac{d|q_i(t)|}{dt} = -\infty, \quad i \in \{1, 2, 3\}. \quad (60)$$

Now, a contradiction between (54) and (60) happens. Thereby, (50) is impossible, i.e., there exist three constants $\sigma_{1i} > 0$, $i = 1, 2, 3$, such that

$$|q_i(t)| \leq k_{1i}(t) - \sigma_{1i} < k_{1i}(t), \quad i = 1, 2, 3, \quad t < \hat{t}. \quad (61)$$

Case 2: Proceed with supposing (51) for verification. It begins with giving the error dynamics by (7) and (29):

$$\dot{e}(t) = F(\omega) + G_0 M u(t) + E(t) - \dot{\alpha}(t). \quad (62)$$

Due to the inputs coupling of (62), the independent analysis of $e_i(t)$, $i = 1, 2, 3$, is difficult. Instead, a unified analysis is adopted. To this end, construct the following BLF:

$$V(t) = \frac{1}{\pi} \eta_2(t)^T \eta_2(t), \quad (63)$$

and there is

$$\dot{V}(t) = \frac{2}{\pi} \eta_2(t)^T \dot{\eta}_2(t). \quad (64)$$

$$\dot{G}_1^{-1}(q) = \frac{1}{2} \begin{bmatrix} 2q_0\dot{q}_0 + 2q_1\dot{q}_1 & \dot{q}_0q_3 + q_0\dot{q}_3 + q_1\dot{q}_2 + \dot{q}_1q_2 & \dot{q}_1q_3 + q_1\dot{q}_3 - \dot{q}_0q_2 - q_0\dot{q}_2 \\ q_1\dot{q}_2 + \dot{q}_1q_2 - \dot{q}_0q_3 - q_0\dot{q}_3 & 2q_0\dot{q}_0 + 2q_2\dot{q}_2 & \dot{q}_0q_1 + q_0\dot{q}_1 + q_2\dot{q}_3 + \dot{q}_2q_3 \\ q_1\dot{q}_3 + q_1\dot{q}_3 + \dot{q}_0q_2 + q_0\dot{q}_2 & q_2\dot{q}_3 + \dot{q}_2q_3 - \dot{q}_0q_1 - q_0\dot{q}_1 & 2q_0\dot{q}_0 + 2q_3\dot{q}_3 \end{bmatrix}. \quad (69)$$

Herein, the differential equation of $\eta_2(t)$ is obtained by differentiating (32) via (33) and (34) as follows

$$\dot{\eta}_2(t) = \frac{\pi}{2} \beta(t)(\dot{e}(t) - h(t)), \quad (65)$$

where

$$h(t) = \left[\frac{\dot{k}_{21}(t)e_1(t)}{k_{21}(t)}, \frac{\dot{k}_{22}(t)e_2(t)}{k_{22}(t)}, \frac{\dot{k}_{23}(t)e_3(t)}{k_{23}(t)} \right]^T. \quad (66)$$

Accordingly, the complete expression of $\dot{V}(t)$ is derived by substituting (65) with (62) into (64) as follows

$$\dot{V}(t) = \eta_2^T(t)\beta(t)(F(\omega) + G_0Mu(t) + E(t) - \dot{\alpha}(t) - h(t)), \quad (67)$$

where the derivative of $\alpha(t)$ in (28) is computed by (5) and (26) as follows

$$\dot{\alpha}(t) = -c_2\dot{G}_1^{-1}(q)\eta_1(t) - c_2G_1^{-1}(q)\dot{\eta}_1(t), \quad (68)$$

with $\dot{G}_1^{-1}(q)$ given by (69), as shown at the top of the page, and

$$\dot{\eta}_{1i}(t) = \frac{\pi}{2} \frac{1}{k_{1i}(t)} \frac{1}{\cos^2\left(\frac{\pi}{2} \cdot \frac{q_i(t)}{k_{1i}(t)}\right)} \left(\dot{q}_i(t) - \frac{\dot{k}_{1i}(t)}{k_{1i}(t)} q_i(t) \right), \quad (70)$$

for $i = 1, 2, 3$. Substituting the designed control inputs in (35) into the dynamical equation in (67) yields

$$\begin{aligned} \dot{V}(t) &= \eta_2^T(t)\beta(t)H(X, t) \\ &\quad - c_2\eta_2^T(t)\beta(t)G_0MG_0^T\beta(t)\eta_2(t), \end{aligned} \quad (71)$$

where

$$H(X, t) = F(\omega) + E(t) - \dot{\alpha}(t) - h(t). \quad (72)$$

Proceed with scaling (71). On the one hand, it follows from Lemma 2 that

$$\eta_2^T(t)\beta(t)G_0MG_0^T\beta(t)\eta_2(t) \geq 0.5g \|\beta(t)\eta_2(t)\|^2. \quad (73)$$

Then, substituting (73) into (71) gives

$$\dot{V}(t) \leq \|H(X, t)\| \cdot \|\beta(t)\eta_2(t)\| - b \|\beta(t)\eta_2(t)\|^2, \quad (74)$$

where $b = 0.5c_2g$. On the other hand, discuss the boundedness of $\|H(X, t)\|$ on $[0, \hat{t})$ as follows. Due to (61), it follows from (26) that $\|\eta_{1i}(t)\| < \infty$, $t < \hat{t}$, $i = 1, 2, 3$. According to (16), there is $\det(G_1^{-1}(q)) = \frac{2}{q_0}$, which together with (25) ensures $\|G_1^{-1}(q)\| < \infty$, $t < \hat{t}$. Thereby, from (28), we know $\|\alpha(t)\| < \infty$, $t < \hat{t}$. Substituting it and (52) into (55), there holds $\|\omega(t)\| < \infty$ on $[0, \hat{t})$. Hence, $\|F(\omega)\|$ in (7) is bounded over $[0, \hat{t})$, due to the continuity of $F(\cdot)$. Property 1 shows the boundedness of $q_i(t)$ on $[0, \hat{t})$, $i = 0, 1, 2, 3$. Moreover, (1) shows $\|\dot{q}_i(t)\| < \infty$, $t < \hat{t}$, $i = 0, 1, 2, 3$. Therefore, it is obtained from (69) that $\|\dot{G}_1^{-1}(q)\|$ is bounded on $[0, \hat{t})$. One sees from (30) that $k_{2i}(t)$, $\dot{k}_{2i}(t)$ and $\frac{1}{k_{2i}(t)}$ are bounded on $[0, \hat{t})$. This along with (52) ensures (66) that $\|h(t)\| < \infty$, $t < \hat{t}$. Furthermore, owing to (61), the above facts guarantee that $\dot{\eta}_{1i}(t)$ in (70) is bounded for $t < \hat{t}$, $i = 1, 2, 3$. Accordingly, the same remains for $\|\dot{\alpha}(t)\|$ in (68) for $t < \hat{t}$. Recall (7) and Assumption 1, and we have $\|E(t)\| < \infty$.

The above facts enable us to conclude the boundedness of $\|H(X, t)\|$ in (72). Then, we let

$$a := \sup_{t \in [0, \hat{t})} \|H(X, t)\|. \quad (75)$$

Accordingly, (74) is further bounded by

$$\begin{aligned} \dot{V}(t) &\leq a \|\beta(t)\eta_2(t)\| - b \|\beta(t)\eta_2(t)\|^2 \\ &= \|\beta(t)\eta_2(t)\| (a - b \|\beta(t)\eta_2(t)\|), \quad t < \hat{t}, \end{aligned}$$

which shows $\dot{V}(t) < 0$ as $\|\beta(t)\eta_2(t)\| > \frac{a}{b}$ for $t < \hat{t}$. Based on Lemma 3, one further has $\dot{V}(t) < 0$, if $\|\eta_2(t)\| > \frac{ap}{b}$ for $t < \hat{t}$. This means by (63) that $\|\eta_2(t)\|$ decreases once $\|\eta_2(t)\| > \frac{ap}{b}$ for $t < \hat{t}$. As a result,

$$\|\eta_2(t)\| \leq \max \left\{ \|\eta_2(0)\|, \frac{ap}{b} \right\}, \quad t < \hat{t}. \quad (76)$$

Recall the barrier functions in (32) and (34) as well as the continuity of $e_i(t)$ and $k_{2i}(t)$, $i = 1, 2, 3$, under (52). Then, the boundedness shown in (76) implies that $e_i(t)$ keeps away from $k_{2i}(t)$ over $[0, \hat{t})$ for each $i = 1, 2, 3$. Otherwise, $V(t)$ defined in (63) tends to infinity. Hence, there exist three constants, $\sigma_{2i} > 0$, $i = 1, 2, 3$, so that

$$|e_i(t)| \leq k_{2i}(t) - \sigma_{2i} < k_{2i}(t), \quad t < \hat{t}, \quad i = 1, 2, 3. \quad (77)$$

Obviously, (50) and (51) contradict (61) and (77), respectively. Consequently, (50) and (51) are invalid. Instead,

$$\begin{cases} |q_i(t)| \leq k_{1i}(t) - \sigma_{1i} < k_{1i}(t), & t \geq 0, \quad i = 1, 2, 3, \\ |e_i(t)| \leq k_{1i}(t) - \sigma_{2i} < k_{1i}(t), & t \geq 0, \quad i = 1, 2, 3. \end{cases} \quad (78)$$

Apparently, the prescribed closed-loop performance in (49) holds, whenever the actuator failures take place. Besides, it is demonstrated that the constrained variables evolve without boundary contacts, which is crucial for the boundedness analysis of other signals below.

The rest of the signals in the control system include the intermediate control signals, $\alpha_i(t)$, $i = 1, 2, 3$, the control signal, $u(t)$, and the angular velocities, $\omega_i(t)$, $i = 1, 2, 3$. Under (78), it follows from (26), (32) and (33) that $\eta_{1i}(t) \in L^\infty$, $\eta_{2i}(t) \in L^\infty$ and $\beta_i(t) \in L^\infty$, $i = 1, 2, 3$. They guarantee the boundedness of $\alpha_i(t)$ in (28) and $\|u(t)\|$ in (35), $i = 1, 2, 3$. Further, from (55), we know that $\omega_i(t)$ in (29) is bounded, $i = 1, 2, 3$.

REFERENCES

- [1] A. G. Stove, M. S. Gashinova, S. Hristov, and M. Cherniakov, "Passive maritime surveillance using satellite communication signals," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 53, no. 6, pp. 2987–2997, Dec. 2017.
- [2] G. S. Percivall, N. S. Alameh, H. Caumont, K. L. Moe, and J. D. Evans, "Improving disaster management using Earth observations—GEOSS and CEOS activities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 3, pp. 1368–1375, Jun. 2013.
- [3] T. Hadas and T. Hobiger, "Benefits of using Galileo for real-time GNSS meteorology," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 10, pp. 1756–1760, Oct. 2021.

- [4] W. Jiang, D. Liu, B. Cai, C. Rizos, J. Wang, and W. Shangguan, "A fault-tolerant tightly coupled GNSS/INS/OVS integration vehicle navigation system based on an FDP algorithm," *IEEE Trans. Veh. Technol.*, vol. 68, no. 7, pp. 6365–6378, Jul. 2019.
- [5] C. Yue, F. Wu, F. Wang, X. Cao, Q. Shen, and K. D. Kumar, "Two parallel single-gimbal control moment gyros actuated spacecraft attitude maneuver," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 56, no. 5, pp. 4112–4123, Oct. 2020.
- [6] J. R. Wertz, *Spacecraft Attitude Determination and Control*, vol. 73. Cham, Switzerland: Springer, 2012.
- [7] L. Sun and Z. Zheng, "Disturbance-observer-based robust backstepping attitude stabilization of spacecraft under input saturation and measurement uncertainty," *IEEE Trans. Ind. Electron.*, vol. 64, no. 10, pp. 7994–8002, Oct. 2017.
- [8] Q. Shen, D. Wang, S. Zhu, and E. K. Poh, "Integral-type sliding mode fault-tolerant control for attitude stabilization of spacecraft," *IEEE Trans. Control Syst. Technol.*, vol. 23, no. 3, pp. 1131–1138, May 2015.
- [9] Y. Liu, B. Jiang, J. Lu, J. Cao, and G. Lu, "Event-triggered sliding mode control for attitude stabilization of a rigid spacecraft," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 50, no. 9, pp. 3290–3299, Sep. 2020.
- [10] S. M. Amrr, J. P. Srivastava, and M. Nabi, "Robust attitude stabilization of spacecraft under constrained network with hysteresis quantizer," *IEEE J. Miniaturization Air Space Syst.*, vol. 2, no. 3, pp. 129–139, Sep. 2021.
- [11] S. M. Amrr, M. Nabi, U. M. Al-Saggaf, and M. Bettayeb, "Attitude control of spacecraft under restricted network using robust event-triggered optimized strategy," *IEEE J. Miniaturization Air Space Syst.*, vol. 2, no. 4, pp. 179–188, Dec. 2021.
- [12] Z. Zhu, Y. Xia, and M. Fu, "Adaptive sliding mode control for attitude stabilization with actuator saturation," *IEEE Trans. Ind. Electron.*, vol. 58, no. 10, pp. 4898–4907, Oct. 2011.
- [13] Z. Guo, Z. Wang, S. Ding, S. Li, and X. Wang, "Global finite-time stabilization of spacecraft attitude with disturbances via continuous nonlinear controller," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 3, pp. 2608–2620, Mar. 2023.
- [14] Y.-Y. Wu and H.-J. Sun, "Attitude tracking control with constraints for rigid spacecraft based on control barrier Lyapunov functions," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 3, pp. 2053–2062, Jun. 2022.
- [15] Q. Chen, S. Xie, M. Sun, and X. He, "Adaptive nonsingular fixed-time attitude stabilization of uncertain spacecraft," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 54, no. 6, pp. 2937–2950, Dec. 2018.
- [16] S. Xie and Q. Chen, "Adaptive nonsingular predefined-time control for attitude stabilization of rigid spacecrafts," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 69, no. 1, pp. 189–193, Jan. 2022.
- [17] Y. Wang, S. Tang, J. Guo, X. Wang, and C. Liu, "Fuzzy-logic-based fixed-time geometric backstepping control on SO(3) for spacecraft attitude tracking," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 55, no. 6, pp. 2938–2950, Dec. 2019.
- [18] Q. Hu, X. Shao, and L. Guo, "Adaptive fault-tolerant attitude tracking control of spacecraft with prescribed performance," *IEEE/ASME Trans. Mechatronics*, vol. 23, no. 1, pp. 331–341, Feb. 2018.
- [19] Y. Hu, Y. Geng, B. Wu, and D. Wang, "Model-free prescribed performance control for spacecraft attitude tracking," *IEEE Trans. Control Syst. Technol.*, vol. 29, no. 1, pp. 165–179, Jan. 2021.
- [20] H. Zhang, Z. Chen, B. Xiao, and B. Li, "Fast fixed-time attitude tracking control of spacecraft with prescribed performance," *Int. J. Robust Nonlinear Control*, vol. 33, no. 10, pp. 5229–5245, Jul. 2023.
- [21] M. Golestani, S. M. Esmailzadeh, and S. Mobayen, "Fixed-time control for high-precision attitude stabilization of flexible spacecraft," *Eur. J. Control*, vol. 57, pp. 222–231, Jan. 2021.
- [22] B. Xiao, H. Zhang, Z. Chen, and L. Cao, "Fixed-time fault-tolerant optimal attitude control of spacecraft with performance constraint via reinforcement learning," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 6, pp. 7715–7724, Jun. 2023.
- [23] C. Zhang, G. Ma, Y. Sun, and C. Li, "Observer-based prescribed performance attitude control for flexible spacecraft with actuator saturation," *ISA Trans.*, vol. 89, pp. 84–95, Jun. 2019.
- [24] C. Yue et al., "A systematic method for constrained attitude control under input saturation," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 5, pp. 6005–6015, May 2023.
- [25] C. Yue, F. Wang, X. Cao, Q. Shen, and X. Chen, "Robust fault-tolerant attitude tracking with guaranteed prescribed performance," *J. Franklin Inst.*, vol. 357, no. 1, pp. 229–253, Jan. 2020.
- [26] C. P. Bechlioulis and G. A. Rovithakis, "A low-complexity global approximation-free control scheme with prescribed performance for unknown pure feedback systems," *Automatica*, vol. 50, no. 4, pp. 1217–1226, 2014.
- [27] N. Zhou, X. Cheng, Y. Xia, and Y. Liu, "Fully adaptive-gain-based intelligent failure-tolerant control for spacecraft attitude stabilization under actuator saturation," *IEEE Trans. Cybern.*, vol. 52, no. 1, pp. 344–356, Jan. 2022.
- [28] B. Li, Q. Hu, G. Ma, and Y. Yang, "Fault-tolerant attitude stabilization incorporating closed-loop control allocation under actuator failure," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 55, no. 4, pp. 1989–2000, Aug. 2019.
- [29] B. Li, Q. Hu, Y. Yu, and G. Ma, "Observer-based fault-tolerant attitude control for rigid spacecraft," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 53, no. 5, pp. 2572–2582, Oct. 2017.
- [30] G. Sun, S. Xu, and Z. Li, "Finite-time fuzzy sampled-data control for nonlinear flexible spacecraft with stochastic actuator failures," *IEEE Trans. Ind. Electron.*, vol. 64, no. 5, pp. 3851–3861, May 2017.
- [31] Q. Shen, D. Wang, S. Zhu, and K. Poh, "Finite-time fault-tolerant attitude stabilization for spacecraft with actuator saturation," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 51, no. 3, pp. 2390–2405, Jul. 2015.
- [32] S. Gao, X. Liu, Y. Jing, and G. M. Dimirovski, "A novel finite-time prescribed performance control scheme for spacecraft attitude tracking," *Aerosp. Sci. Technol.*, vol. 118, Nov. 2021, Art. no. 107044.
- [33] X. Shao, Q. Hu, Y. Shi, and B. Jiang, "Fault-tolerant prescribed performance attitude tracking control for spacecraft under input saturation," *IEEE Trans. Control Syst. Technol.*, vol. 28, no. 2, pp. 574–582, Mar. 2020.
- [34] X. Huang, Z. Dong, K. Zhang, F. Zhang, and L. Zhang, "Approximation-free attitude fault-tolerant tracking control of rigid spacecraft with global stability and appointed accuracy," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 5, pp. 7035–7046, May 2023.
- [35] M. Qian, Y. Shi, Z. Gao, and X. Zhang, "Integrated fault tolerant tracking control for rigid spacecraft using fractional order sliding mode technique," *J. Franklin Inst.*, vol. 357, no. 15, pp. 10557–10583, Oct. 2020.
- [36] M. Qian, C. Sun, B. Jiang, R. Wang, and J. Shi, "Event-based adaptive fault tolerant control and collision avoidance of wheel mobile robots with communication limits," *IEEE Trans. Ind. Electron.*, vol. 71, no. 11, pp. 14832–14841, Nov. 2024.
- [37] J. Lei, T. Meng, K. Wang, W. Wang, and S. Sun, "Composite event-triggered intermittent attitude control of spacecraft," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 2, pp. 1612–1627, Apr. 2024.
- [38] J.-X. Zhang, J. Ding, and T. Chai, "Fault-tolerant prescribed performance control of wheeled mobile robots: A mixed-gain adaption approach," *IEEE Trans. Autom. Control*, vol. 69, no. 8, pp. 5500–5507, Aug. 2024.
- [39] Q. Shen, C. Yue, C. H. Goh, B. Wu, and D. Wang, "Rigid-body attitude stabilization with attitude and angular rate constraints," *Automatica*, vol. 90, pp. 157–163, Apr. 2018.
- [40] Y. Song, X. Huang, and C. Wen, "Robust adaptive fault-tolerant PID control of MIMO nonlinear systems with unknown control direction," *IEEE Trans. Ind. Electron.*, vol. 64, no. 6, pp. 4876–4884, Jun. 2017.
- [41] Y. Song, B. Zhang, and K. Zhao, "Indirect neuroadaptive control of unknown MIMO systems tracking uncertain target under sensor failures," *Automatica*, vol. 77, pp. 103–111, Mar. 2017.
- [42] H. Xie, J.-X. Zhang, Y. Jing, G. Marko Dimirovski, and J. Chen, "Self-adjustable performance-based adaptive tracking control of uncertain nonlinear systems," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 750–764, 2025.
- [43] J. X. Zhang and G. H. Yang, "Low-complexity tracking control of strict-feedback systems with unknown control directions," *IEEE Trans. Autom. Control*, vol. 64, no. 12, pp. 5175–5182, Dec. 2019.
- [44] X. Gao, E. Cui, D. Yang, Z. Tan, and J. Sun, "Adaptive displacement constraint control with predefined performance for active magnetic bearings," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 883–894, 2025.
- [45] J. Liang, Y. Chen, Y. Wu, Z. Miao, H. Zhang, and Y. Wang, "Adaptive prescribed performance control of unmanned aerial manipulator with disturbances," *IEEE Trans. Autom. Sci. Eng.*, vol. 20, no. 3, pp. 1804–1814, Jul. 2023.

- [46] J.-X. Zhang, J. Ding, and T. Chai, "Cyclic performance monitoring-based fault-tolerant funnel control of unknown nonlinear systems with actuator failures," *IEEE Trans. Autom. Control*, early access, Mar. 11, 2025, doi: [10.1109/TAC.2025.3550352](https://doi.org/10.1109/TAC.2025.3550352).
- [47] J.-X. Zhang and T. Chai, "Proportional-integral funnel control of unknown lower-triangular nonlinear systems," *IEEE Trans. Autom. Control*, vol. 69, no. 3, pp. 1921–1927, Mar. 2024.
- [48] J.-X. Zhang, J.-G. Song, Q. Chen, and F. Yang, "Mixed-gain adaption-based fault-tolerant funnel control of robotic manipulators with unknown dynamics and sensor faults," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 3044–3055, 2025.



Jin-Xi Zhang (Member, IEEE) received the B.S. degree in automation and the Ph.D. degree in control theory and control engineering from Northeastern University, Shenyang, China, in 2014 and 2020, respectively.

From 2019 to 2020, he was a Research Fellow with the Institute for Intelligent Systems, Faculty of Engineering and the Built Environment, University of Johannesburg, Johannesburg, South Africa. He is currently a Distinguished Associate Professor with the State Key Laboratory of Synthetical Automation for Process Industries, Northeastern University. His current research interests include nonlinear control, intelligent control, prescribed performance control, multiagent systems, fault diagnosis and fault-tolerant control, and image processing.

Dr. Zhang is also an Associate Editor of *International Journal of Fuzzy Systems* and *Journal of Automation and Intelligence*.



Yun-Qi Liu received the B.S. degree in mechatronic engineering from Changchun University of Technology, Changchun, China, in 2021. He is currently pursuing the M.S. degree in control theory and control engineering with Northeastern University, Shenyang, China.

His current research interests include nonlinear systems, fault-tolerant control, and prescribed performance control.



Tianyou Chai (Life Fellow, IEEE) received the Ph.D. degree in control theory and engineering from Northeastern University, Shenyang, China, in 1985.

He became a Professor with Northeastern University in 1988. He is currently the Founder and the Director of the Center of Automation, which became the National Engineering and Technology Research Center and the State Key Laboratory of Synthetical Automation for Process Industries. He has served as the Director of the Department of Information Science of the National Natural Science Foundation

of China from 2010 to 2018. He has published 297 peer-reviewed international journal articles. His current research interests include modeling, control, optimization, and integrated automation of complex industrial processes. He is a member of Chinese Academy of Engineering and an IFAC Fellow. His article titled *Hybrid intelligent control for optimal operation of shaft furnace roasting process* was selected as one of three best papers for the Control Engineering Practice Paper Prize from 2011 to 2013. He has developed control technologies with applications to various industrial processes. For his contributions, he has won five prestigious awards of the National Natural Science, the National Science and Technology Progress, the National Technological Innovation, the 2007 Industry Award for Excellence in Transitional Control Research from the IEEE Multiple-Conference on Systems and Control, and the 2017 Wook Hyun Kwon Education Award from Asian Control Association.